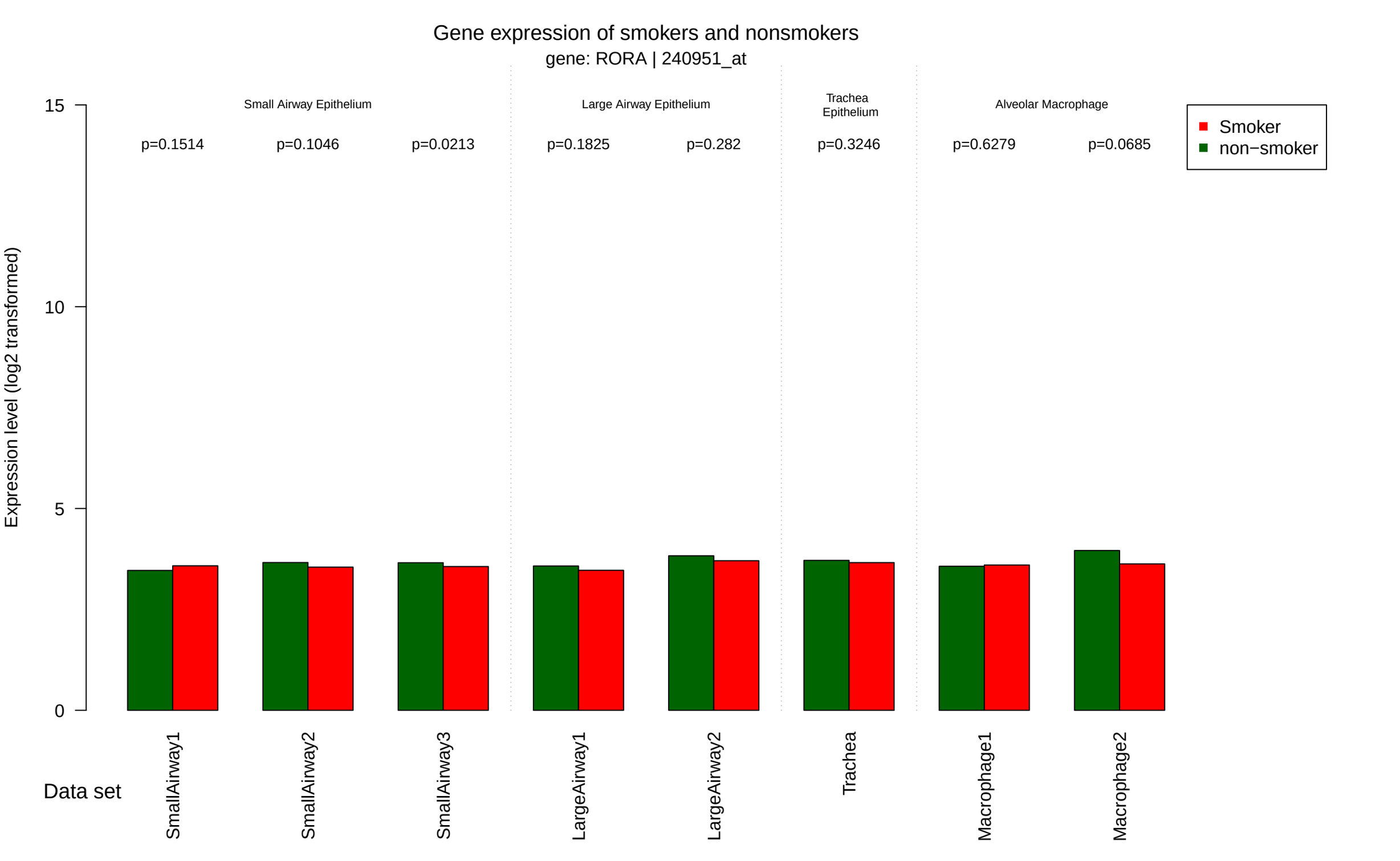
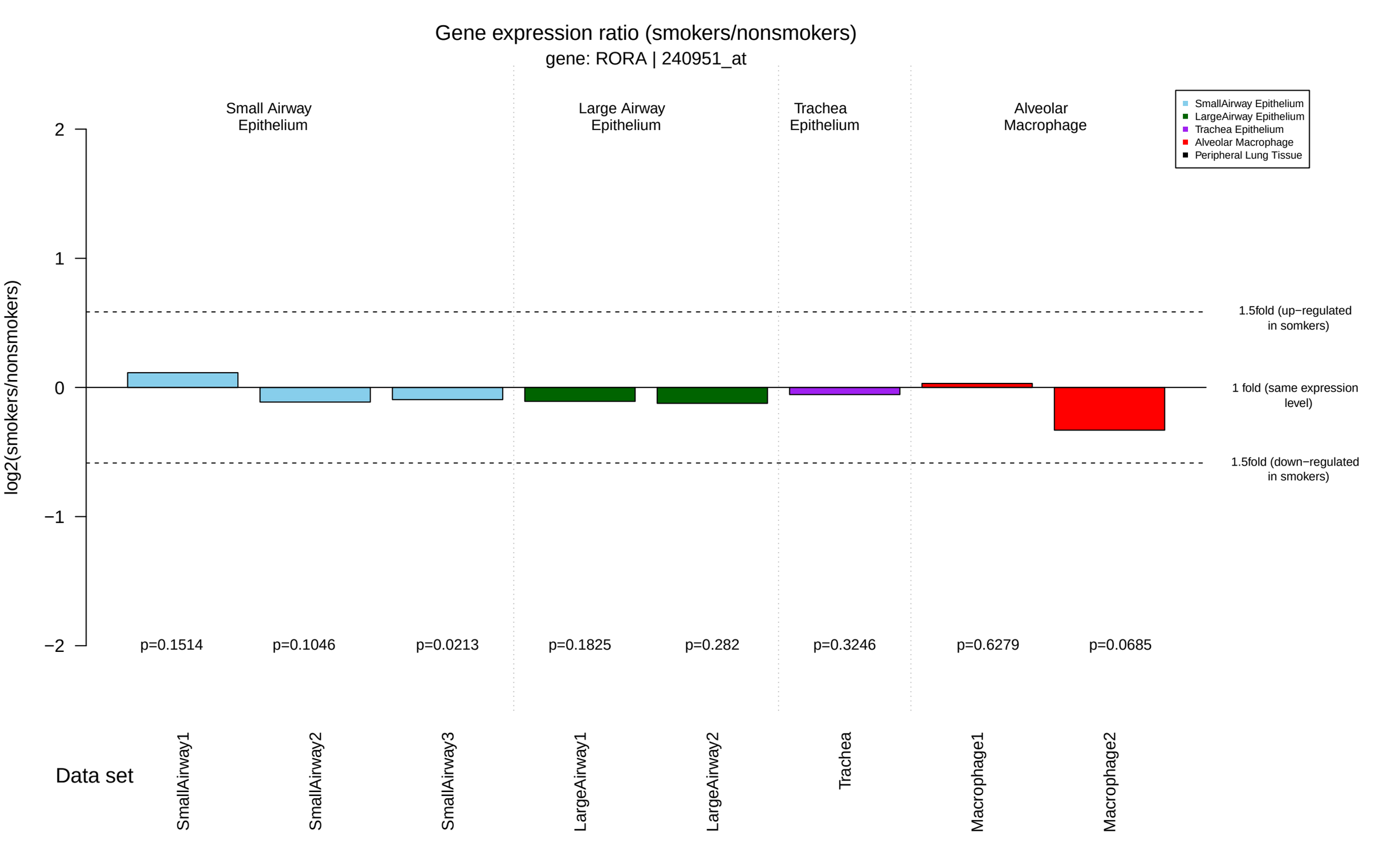
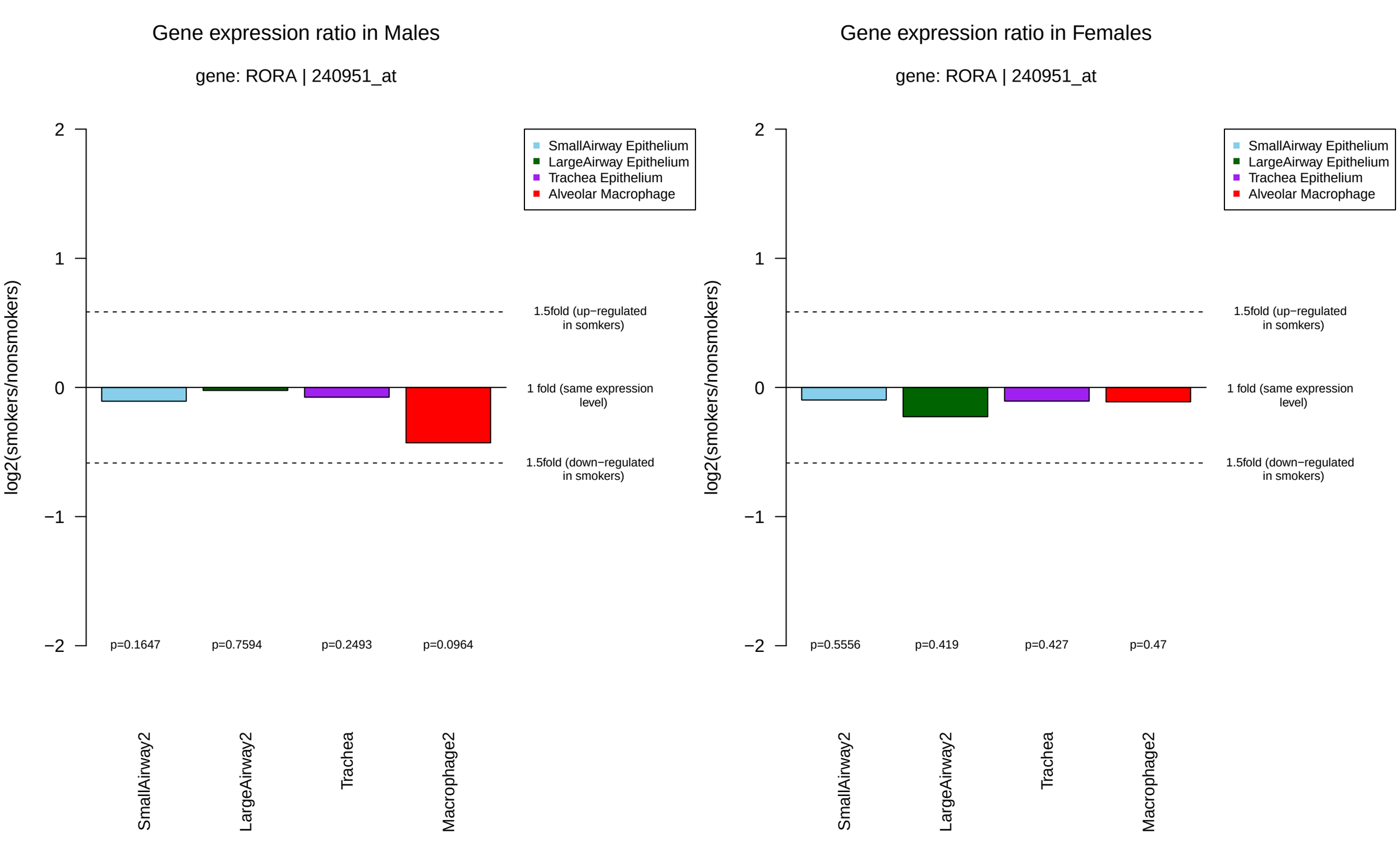
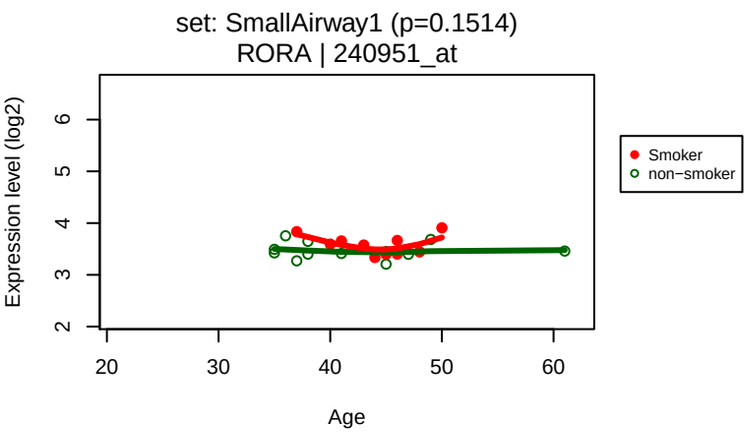
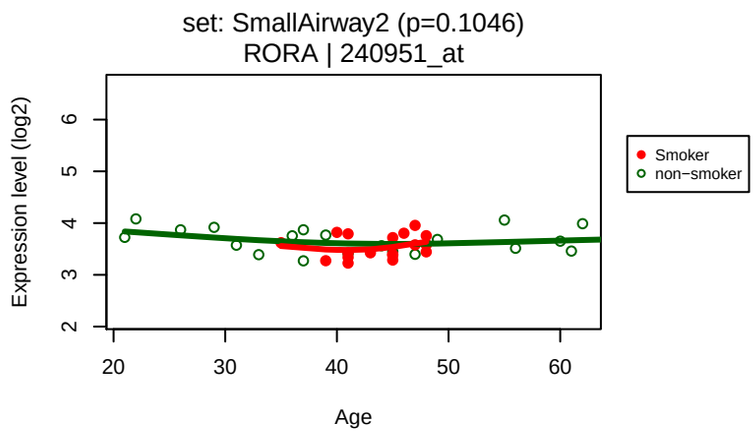
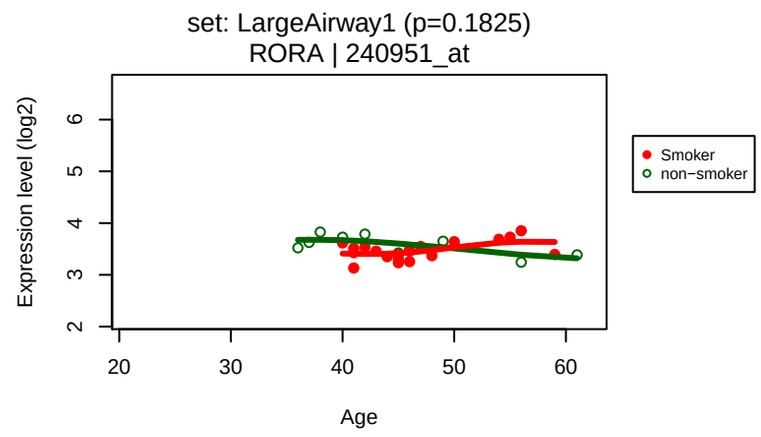

Gene expression of smokers and nonsmokers
gene: RORA | 240951_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1514
p=0.1046
p=0.0213
p=0.1825
p=0.282
p=0.3246
p=0.6279
p=0.0685
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

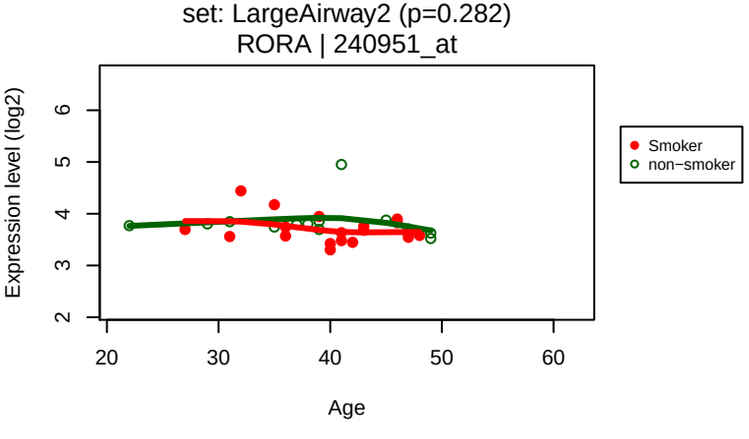

Gene expression ratio (smokers/nonsmokers)
gene: RORA | 240951_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1514
p=0.1046
p=0.0213
p=0.1825
p=0.282
p=0.3246
p=0.6279
p=0.0685
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

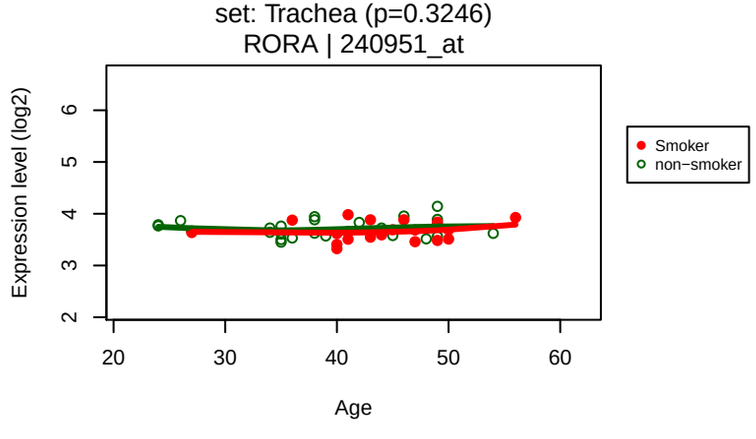

Gene expression ratio in Males
gene: RORA | 240951_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1647
p=0.7594
p=0.2493
p=0.0964
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RORA | 240951_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5556
p=0.419
p=0.427
p=0.47
SmallAirway2
LargeAirway2
Trachea
Macrophage2

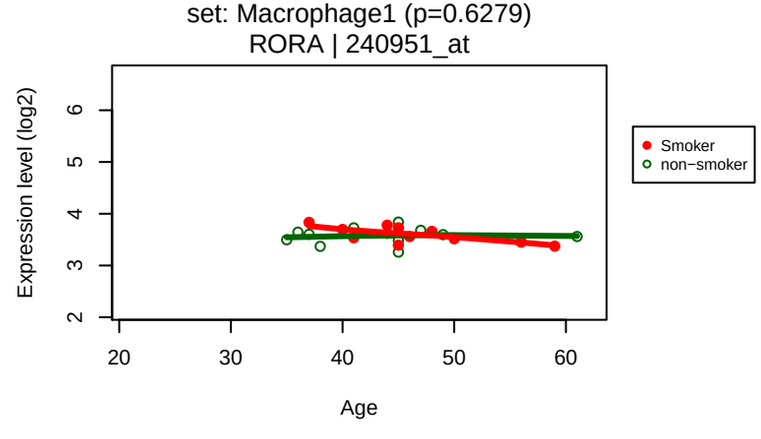

set: SmallAirway1 (p=0.1514)
RORA | 240951_at
Expression level (log2)
Age
Smoker
non−smoker

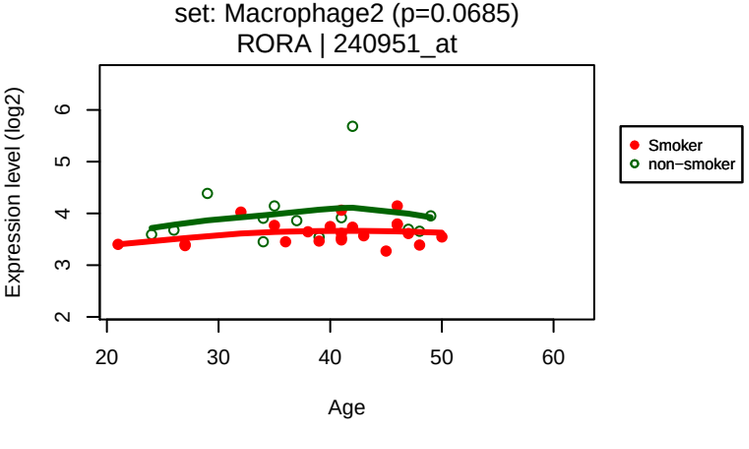

set: SmallAirway2 (p=0.1046)
RORA | 240951_at
Expression level (log2)
Age
Smoker
non−smoker

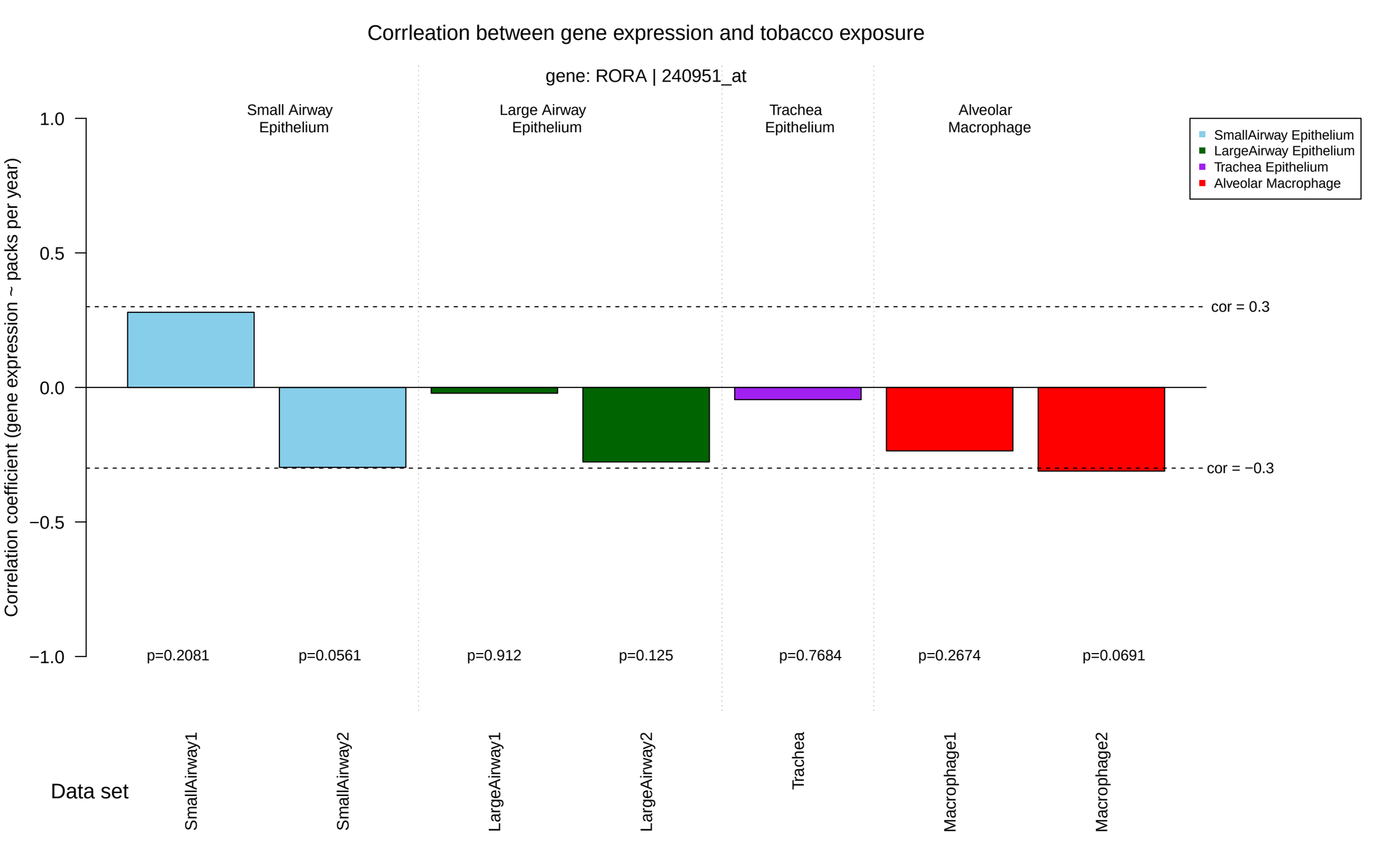

set: LargeAirway1 (p=0.1825)
RORA | 240951_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.282)
RORA | 240951_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3246)
RORA | 240951_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6279)
RORA | 240951_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0685)
RORA | 240951_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RORA | 240951_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2081
p=0.0561
p=0.912
p=0.125
p=0.7684
p=0.2674
p=0.0691
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2